

PETALNEX PVT LTD

Internship — Daily Progress Report

Intern Name	Muhammad Bin Waseem
Date	03 / 09 / 2026

Daily Report Fields

Today I Learned	Learned how to integrate AI, RAG, structured output validation, fallback handling, human escalation, and notifications into an existing deterministic automation workflow.
Today I Built / Practiced	Integrated the AI/RAG layer into the Company Knowledge AI Assistant. Connected the existing company knowledge base with AI-generated responses, source attribution, structured output, safe fallback responses, and escalation for questions that could not be answered from the knowledge base.
Technologies / Tools Used	n8n, RAG, AI/LLM, Supabase Vector Store, OpenAI, structured output validation, human-in-the-loop, notifications, and workflow automation.
Problem(s) Faced	Some questions were not covered by the company knowledge base and therefore could not be answered reliably by the AI.
How I Solved Them	Implemented a safe knowledge-base fallback and escalation mechanism so unsupported questions are not answered with fabricated information and can instead be reviewed by a human.
Task Status	Completed
GitHub / Workflow Link	https://github.com/MuhammadBinWaseemm/AI-Automation-Internship-Pentalnex-Pvt.-Ltd/tree/main/Final%20Capstone%20Project/Development-AI,%20Integration%20&%20Notifications
Plan for Next Day	Going to do a lot of testing on my final project.